Qatra قطرة

Technical and Strategic Dossier

Blood-donation matching for Algeria

Families ask, associations vouch, donors answer.

Every figure in this document was taken from the repository, from the live database, or from a named public source. Test results are from runs on the date below. Nothing here is aspirational unless it appears in Section **??**, which exists to say what is missing.

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1. Executive summary

Algeria is the best country in Africa at collecting blood and still runs short of it. The Agence Nationale du Sang reports roughly **677 000 bags** from over **800 000 donors**, a rate of **13.98 donors per 1 000 inhabitants** — the highest on the continent for a decade — and describes its own problem not as a shortage of willing people but as *"ce déséquilibre entre l'offre et la demande"*: a timing failure. Only **22% of donors give regularly**, which the agency calls *"relativement faible"*, and its stated request is that citizens **register on regular-donor lists**.

Qatra answers that request. It knows the date every donor becomes eligible again, and it reaches the compatible ones the moment somebody nearby needs their type.

Three things distinguish it from what already exists:

1. **Requests are vouched for.** A verified association in the same wilaya can put its name to a request. The market-leading competitor states in its own store listing that it verifies nothing.
2. **Health data is handled lawfully.** Consent is recorded per purpose with the version of the text shown; a donor's phone number is masked until deliberately opened, and every opening is logged in a record the donor can read and nobody can delete.
3. **The loop closes.** Post, notify, respond, and the family sees who is coming. A registry answers *who could give*. Qatra answers *who is coming, right now*.

2. The problem, with evidence

2.1 Algeria does not have a donor shortage

Figure	Meaning
13.98 per 1 000	Donation rate. First in Africa for around a decade.
677 000 bags	Collected, from 800 000+ donors; up 6.75% since 2023.
22%	**Share of donors who give regularly.** ANS: *"relativement faible"*.
58	Wilayas. Blood is coordinated locally; a donor two provinces away cannot help today.

The ANS names the summer as its pressure point — road-accident season — and has run joint campaigns with the national police and Algérie Télécom to cover it. Its ask of the public is continuity, not volume: *"participer à l'alimentation continue de la banque du sang"*.

2.2 Which means the problem is reach, not recruitment

Seventy-eight per cent of Algerian donors give once and are never effectively called again. Nothing in the existing landscape knows when a given donor became eligible again, or tells them that somebody two kilometres away needs exactly their type this evening. That is the gap this product occupies.

3. Competitive landscape

Install counts and ratings retrieved from Google Play on 24 August 2026. Platform claims verified against the operators' own sites on the same date.

Product	Installs	Rating	Observation
بنك الدم الجزائري (Nabil_Soft)	10K+	4.7 / 130	Market leader. **Contains ads.**
بنك الدم الجزائري (SangDz)	1K+	—	Solo developer based in France.
Sadaka صدقة	1K+	—	—
Nabdz	—	—	Best-designed. 2 requests, 3 donors live.
ONSC / Ministry platforms	—	—	Announced June 2025. No public address.

Approximately **12 000 installs across every blood application in Algeria**, in a country of 45 million with the continent's highest donation rate. The field is not crowded; it is empty.

3.1 The announced national platforms have no front door

The National Observatory of Civil Society launched a donor platform with the Algerian Red Crescent on 14 June 2025; the Ministry of Health announced its own three days later. Fourteen months on:

- `marsad.dz` is the Observatory's institutional site — governance, compliance, tenders. It is actively maintained, with content through April 2026, and carries no donor platform.
- Neither announcement names a URL, an application, or any route by which a citizen reaches the system.
- The Ministry's own site lists five other digital services it does ship — a pharmacy system, a health map, a professional space, a training registry — and no blood platform.

A launch with ministers, a telecom sponsor and no discoverable product is a press event. That may change, and the strategic response is not to compete with it (Section **??**).

3.2 Three findings about the market leader

It carries advertising. بنك الدم الجزائري is listed as *Contains ads*. Its developer's other titles are *Easy Hairstyles for Girls*, *PermisDZ*, and several baccalaureate revision apps. Blood donation is the sixth item in a portfolio.

Its data-safety declaration states "No data collected." The application collects, by its own description, a donor's name, wilaya, **blood type** and telephone number. Blood type is health data. That declaration is inaccurate on its face.

It disclaims verification in writing:

> *"This application does not provide any medical services or health advice and does not represent any medical or governmental entity... The information provided is provided by the users."*

The fraud problem is therefore not an unrecognised gap in the market. It is a term of service.

3.3 And the most serious competitor demonstrates the same emptiness

Nabdz is the best-built of them — 58 wilayas, urgency tiers, a pledge mechanic comparable to Qatra's response counter. Its live public grid currently shows two requests and three donors, one of the requests titled **"Test Patient"**.

4. Regulatory position

This section states the reasoning behind decisions already implemented in the codebase. It is engineering rationale, not legal advice, and the open question in Section **??** *should be settled with counsel or with the ANS before any public launch.*

4.1 Health data — Loi 18-07 and Loi 25-11

A patient's name attached to a blood type, and a donor's telephone number attached to theirs, are health data. Qatra treats them accordingly:

- **Consent is recorded per purpose, with the version of the text shown.** Two purposes exist — `health_data` and `contact_sharing` — each stamped with the wording the user actually read (`health-data-v1`, `contact-sharing-v1`). Consent given against superseded wording stops granting access until the user agrees again.
- **Consent records have no delete policy.** They are the evidence that processing was lawful, and evidence a controller can erase is not evidence.
- **Donor numbers are masked until deliberately opened.** Search returns `05 •• •• •• 56`. A single function returns the whole number, and it writes a row while doing so.
- **Every reveal is logged, and the donor can read the log.** The record names who opened the number, for which donor, on behalf of which association, and when. It cannot be updated or deleted by anyone.
- **Data-subject rights have a route in the product** — export, correction, deletion — rather than an email address on a policy page.

4.2 Who may vouch — Loi 12-06 on associations

Verification publishes an attestation under an association's name. An association is a legal person acting through its statutory representatives, not through whichever member happens to be holding a phone; if a vouched request proves fraudulent, the association carries it. Verification is therefore restricted to association *administrators* in the request's own wilaya, enforced in the database rather than the interface.

Volunteers keep everything else — the request list, donor search, the work of mobilising people. They simply cannot sign the association's name.

"Administrator" is a proxy for "entitled to bind the association". A real committee's mandate structure may be richer, and blood-request vetting is not itself a regulated medical act. This mapping is marked in the migration and should be confirmed with the ANS.

4.3 No payment, ever

Blood donation in Algeria is voluntary and unpaid. Paying donors — in cash, vouchers, credit, priority, or anything exchangeable for them — compromises the safety of the supply, because a donor with a financial motive has a reason to conceal a disqualifying history.

This is a documented invariant at the top of the compliance module and a code-review gate, not a preference. Non-transferable recognition (streaks, certificates) is permitted precisely because it cannot be exchanged.

4.4 Open question: data residency

The production database is hosted in AWS `us-east-1`. Algerian patients' names, blood types and telephone numbers therefore rest on servers outside Algeria. Four files in the repository carry a `TODO(compliance)` marker recording this. It is a launch gate, and it is the question a well-informed panel asks first.

The answer is that nothing prevents it being solved, and this can be checked rather than taken on trust. The backend is standard PostgreSQL. Across all twenty-six migrations there is not a single `create extension` statement, and no dependency on any proprietary service — no hosted-only networking, secrets or storage calls. The one platform-specific thing the schema uses is `auth.uid()`, which belongs to GoTrue, an open-source component that runs the same way in a self-hosted deployment. The client is handed its address at startup by injection rather than reading it from a bundler's environment, so pointing the application at an instance hosted in Algeria is a configuration change and a migration run, not a rewrite.

Two pieces would still need attention on that path, and are named here rather than discovered later: the push-notification worker is a Deno function and would need somewhere to run, and the demonstration OTP provider must be replaced by a real SMS route before any number is trusted.

The distinction worth drawing is between a system that has not yet moved and one that cannot. A product built on a proprietary mobile backend cannot be hosted domestically at all, whatever its owner decides. This one is a Postgres database and a static web application; where it runs is an operational choice that can be made by whoever ends up accountable for the data.

5. How the loop closes

1. **A family posts a request.** Three steps: who needs blood, where, how urgent. The hospital is free text — a family knows their hospital by name, not by an identifier in a directory.
2. **Phone verification interrupts *after* the form, not before.** A plea written at 3am is captured while the person still has the words; the draft is held and posted the moment the number is confirmed. It is never asked for twice.
3. **Compatible donors nearby are notified.** Same wilaya, compatible blood type, eligible now — not inside the 90-day cooldown — and never the person who posted it.
4. **A committee may vouch for it.** The console groups requests by what to do next: needs review, open a long time, already vouched.
5. **A donor responds, and the family sees it.** The count is public so twenty people do not arrive for two units; the identities are not.
6. **A donor who cannot go says so.** Withdrawal is a state change the family sees, not an erasure.

6. Technical architecture

6.1 Stack

Web application	React 18, Vite 6, TypeScript, Tailwind for layout
Mobile shells	Capacitor (Android, iOS) over the same source
Backend	Supabase: PostgreSQL, PostgREST, GoTrue, Edge Functions (Deno)
Notifications	Web Push (VAPID), queued in the database, sent by a worker
Maps	Leaflet over OpenStreetMap
Languages	English, French, Arabic, with full right-to-left support

The core package is framework-agnostic by design: configuration is injected rather than read from the bundler's environment, so a non-Vite host can consume the same logic.

6.2 Security model

Authorisation lives in the database, not the interface. **54 row-level security policies** and **23 `SECURITY DEFINER` functions** decide who may read and write what; the client can only ask.

Three principles are worth stating because they were arrived at by correcting mistakes:

- **One right per predicate.** A single function once gated vouching, patient-record access *and* donor search. Narrowing it for one purpose silently removed a right unrelated to the decision. They are now separate.
- **Two layers, not one.** Revoking a function from `public` does not remove Supabase's separate grants to the `anon` and `authenticated` roles. Several functions were therefore reachable by anonymous callers and stopped only by their own internal guard. The test suite now asserts the grants themselves.
- **Logs that cannot be rewritten.** Consent records and contact reveals have no update or delete policy at all.

6.3 Notifications

Database triggers write to an outbox; an edge function drains it. An HTTP call inside the transaction would make *posting a plea for blood* fail whenever a push service was slow, and a failed send would leave nothing to retry.

Each claim takes a five-minute lease. `SKIP LOCKED` alone separates only simultaneous workers — a second worker a moment later would send the same notification again, and the test suite caught exactly that.

The payload carries blood type and wilaya and nothing else. A push lands on a lock screen that anyone nearby can read, so it says only what the public request list already shows.

6.4 Blood compatibility

The full eight-by-eight red-cell table is written out rather than derived, so a reader can check it against a transfusion chart line by line. Unknown or missing types answer *false*, never *true*: the safe answer to "do we know this is safe?" is no.

The unit tests check the table from both directions against an independently written chart, verify that rhesus is not symmetric, and confirm that a donor whose type is unrecorded is asked rather than refused.

6.5 Verification and testing

Suite	Count	What it covers
`verify:db`	188	RLS against real PostgreSQL, as several users
`test:unit`	14	Blood compatibility both directions; pledge arithmetic
Playwright	48	End-to-end journeys, three languages, two viewports
Migrations	26	Applied in filename order
Screens	24	
Components	41	
Translated strings	388	Each in all three languages, enforced by the type system

A missing translation is a compilation error, not a visible gap: the string table is a TypeScript interface, so a key added in one language and forgotten in another fails the build.

7. What is built, and what is not

State	Capability
Built	Patient/family posts a request; association verifies it
Built	Donor discovery, map, urgency and recency ordering
Built	Donor responds; family sees the count; withdrawal supported
Built	Web push: compatible donors in-wilaya, and the family on a response
Built	Consent, contact-reveal logging, data-subject requests
Built	Blood compatibility with tests
Built	Trilingual EN / FR / AR with right-to-left
Built	Committee invite links: a committee brings the donors it already has
Built	Public demonstration deployment
Open	Demonstration OTP is enabled: any number can be claimed
Open	Data residency (Section **??**)
Open	"Administrator" as proxy for authority to bind an association
Not built	Native push for the Capacitor Android build (needs FCM)
Not built	Android build refreshed against the current application
Not built	Pagination beyond a 200-request cap per wilaya
Not built	Any integration with a national registry or hospital system

8. Positioning

8.1 Not "first"

There is a national registry announced and at least three applications live. Claiming to be first is both false and checkable in thirty seconds, and volunteering the competition first reads as command where being caught by it reads as naivety.

8.2 The line

> *Algeria has a national donor registry. It answers* who could give. *Qatra answers* who is coming, right now, and can this request be trusted — *and it does so to a standard a committee can put its name on.*

8.3 Why the Red Crescent partnership strengthens

The Algerian Red Crescent is already the state's chosen partner in this space, having co-launched the June 2025 platform. A CRA wilaya committee vouching through Qatra is therefore not a rival to the national effort but the local layer it does not have. The compliance architecture is what makes that partnership possible: an institution can only lend its name to a system that handles health data properly.

9. Presenting this

This section is written for the person presenting, not for the reader being presented to. Everything in it is drawn from the sections above; nothing new is claimed here.

9.1 The sentence to open and close with

> **Algeria is the best country in Africa at collecting blood and still runs short of it. That is a timing problem, not a willingness problem — and timing is a software problem.**

Say it first. Say it again at the end. Everything between the two is evidence for it.

9.2 The first thirty seconds

Open with the fact that contradicts the room's assumption. Everyone expects a plea about apathy; the ANS's own numbers say the opposite.

1. **13.98 donors per 1 000 inhabitants.** First in Africa, for about a decade. Algerians give.
2. **Only 22% give regularly.** The agency's own word for this is *"relativement faible"*.
3. **So 78% gave once and were never effectively called again.** Nothing in the country knows the date a given donor became eligible again.

Then the turn: *the ANS's stated request to citizens is to register on regular-donor lists. This is that register, and it makes the call.*

9.3 The demonstration, in this order

Four minutes. Resist showing the whole app; show the loop closing, because that is the thing no competitor has.

	Show	Say
1	A family posts a request	"Three fields. The hospital is free text, because a family knows its hospital by name, not by an identifier."
2	A committee vouches for it	"An association in the same wilaya puts its name to this. The market leader states in its own store listing that it verifies nothing."
3	The badge on the donor's screen	"And it survives a WhatsApp forward — the committee is named in the shared message, so a forwarded plea stays checkable."
4	A donor responds	"Compatible donors in that wilaya were notified. Not everyone: the right blood types, eligible today, never the person who posted it."
5	The progress bar	"Three of four units pledged. The twentieth donor can see the request is covered and go find another one."
6	The invite screen	"And this is how a Red Crescent committee brings the donor list it already has — a link, never an upload. Each donor signs up and consents themselves."

9.4 The argument that matters to a public body

A ministry does not adopt the best application. It adopts the one it *can* adopt. Frame the competition as a question of eligibility rather than of features:

- The market leader has ten thousand installs and **publishes donors' telephone numbers**, while declaring "No data collected" on a listing that collects blood type and phone number. No Algerian public body can put its name to that.
- A product built on a proprietary foreign mobile backend **cannot be hosted in Algeria at all**, whatever its owner decides.
- The announced national platforms have no public front door.

So the claim is not "ours is nicer". It is:

This is the only one of these that could lawfully become national infrastructure.

9.5 Three things to run, rather than describe

Claims are discounted; commands are not. Each of these takes under two minutes.

Command	What it proves in front of them
`npm run verify:db`	188 checks against a real PostgreSQL that is *not* Supabase. This is the answer to "can this be hosted in Algeria" — the schema runs on any Postgres, on any server, in any country.
`npm run test:unit`	The blood-compatibility table, checked in both directions against an independently written chart. The one place where being wrong is a medical error.
`npm test`	48 end-to-end journeys through a real browser, in three languages and two viewports.

9.6 Questions that will be asked, and honest answers

The weak answers are the ones given defensively. Each of these is better answered plainly.

"How many people use it?" *Nobody yet.* Say it without flinching, then say what it is designed for: one wilaya committee with the donors it already has. The invite mechanism exists for exactly that, and it works. A pilot with one committee is the ask, not a claim of traction.

"Where is the data?" Today, AWS `us-east-1` — outside Algeria, and a launch gate rather than a demonstration gate. Then Section **??**: no proprietary extensions, no hosted-only services, the address injected at startup. Offer to run `verify:db` on the spot.

"How do you know a request is real?" Three layers, and none of them is a claim of certainty: the poster's phone number is verified, an association in the same wilaya can vouch under its own name, and the committee sees a plausibility panel before it does. Qatra reduces the cost of a lie; it does not pretend to eliminate it.

"What stops someone harvesting donors' numbers?" Search returns masked numbers. Opening one is a deliberate act, restricted to members of an association in that wilaya, and every opening is written to a log the donor can read and nobody can delete.

"Isn't this just a WhatsApp group?" A chain message is unattributable and never closes. This one carries the vouching committee's name into the forward, and the family sees who is actually coming.

"What if the state builds its own?" It announced platforms and did not ship a front door. And the architecture is deliberately portable — if the right answer is that the state operates this, that is a handover, not a competitor.

"How does it make money?" Be careful and be honest. Donor payment is excluded by law and by design, so there is no consumer revenue here and there should not be. The realistic routes are institutional: tooling paid for by the bodies that coordinate donation, or public funding. **This is not solved, and claiming otherwise in front of people who fund things is the fastest way to lose them.**

"Is phone verification real?" In the demonstration build, no — a fixed code is accepted, because there is no SMS provider connected. It is a flag, it is off in production, and it is listed in Section **??**. Volunteering this before being caught by it is worth more than the point it costs.

9.7 What not to claim

- **Do not claim users, pilots or partnerships that do not exist.** The Red Crescent is the intended partner, not a signed one. One checkable exaggeration discredits every figure in this document.

- **Do not claim lives saved.** The application deliberately removed that number because nothing counts it.
- **Do not promise medical outcomes.** Qatra moves information between people; it does not transfuse anyone.
- **Do not hide the open items.** Section **??** exists so that a panel finds nothing you did not already say. A gap you disclosed is competence; the same gap discovered is a credibility problem.

9.8 The ask

A presentation without a request is a report. Ask for the two things that are actually needed and cannot be built:

1. **One wilaya committee, for one pilot.** Twenty real donors on a real list, for one season. That converts every architectural claim in this document into evidence.
2. **An introduction to the Agence Nationale du Sang.** Two questions need an authoritative answer: who may bind an association, and where the data must live.

10. Sources

- Agence Nationale du Sang, on regular-donor lists and the summer supply–demand imbalance: `https://news.radioalgerie.dz/fr/node/47717`
- ANS collection figures and donation rate: `https://www.algerie360.com/don-de-sang-lagence-nationale-donne-`
- ONSC platform launch: `https://www.dzair-tube.dz/en/algeria-launches-first-digital-blood-donor-platfo`
- Ministry of Health platform: `https://meatechwatch.com/2025/06/18/algeria-launches-digital-platform-to-e`
- National Observatory of Civil Society: `https://marsad.dz`
- Ministry of Health: `https://www.sante.gov.dz`
- بنك الدم الجزائري (Nabil_Soft) on Google Play: `https://play.google.com/store/apps/details?id=com.dz.bank.blood.nabilsoft.bankblooddz`
- بنك الدم الجزائري SangDz on Google Play: `https://play.google.com/store/apps/details?id=com.blood.sangdz`
- Nabdz emergency grid: `https://nabdz.com/en/blood-donation`